

Pandas Joining and merging DataFrame

1. Write a Pandas program to join the two given dataframes along rows and assign all data.

Test Data:

student_data1:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222
4	S5	Kwame Morin	199

student_data2:

	student_id	name	marks
0	S4	Scarlette Fisher	201
1	S5	Carla Williamson	200
2	S6	Dante Morse	198
3	S7	Kaiser William	219
4	S8	Madeeha Preston	201

2. Write a Pandas program to join the two given dataframes along columns and assign all data.

Test Data:

student_data1:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222
4	S5	Kwame Morin	199

student_data2:

	student_id	name	marks
0	S4	Scarlette Fisher	201
1	S5	Carla Williamson	200
2	S6	Dante Morse	198
3	S7	Kaiser William	219
4	S8	Madeeha Preston	201

3. Write a Pandas program to append rows to an existing DataFrame and display the combined data.

Test Data:

student_data1

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222
4	S5	Kwame Morin	199

New Row(s)

student_id	S6
name	Scarlette Fisher
marks	205

dtype: object

4. Write a Pandas program to append a list of dictionaries or series to a existing DataFrame and display the combined data.

Test Data:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222

4 S5 Kwame Morin 199

Dictionary:

```
student_id            S6
name            Scarlett Fisher
marks            205
dtype: object
```

5. Write a Pandas program to join the two given dataframes along rows and merge with another dataframe along the common column id.

Test Data:

student_data1:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222
4	S5	Kwame Morin	199

student_data2:

	student_id	name	marks
0	S4	Scarlette Fisher	201
1	S5	Carla Williamson	200
2	S6	Dante Morse	198
3	S7	Kaiser William	219
4	S8	Madeeha Preston	201

exam_data:

	student_id	exam_id
0	S1	23
1	S2	45
2	S3	12
3	S4	67
4	S5	21
5	S7	55
6	S8	33
7	S9	14

8	S10	56
9	S11	83
10	S12	88
11	S13	12

6. Write a Pandas program to join the two dataframes using the common column of both dataframes.

Test Data:

student_data1:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222
4	S5	Kwame Morin	199

student_data2:

	student_id	name	marks
0	S4	Scarlette Fisher	201
1	S5	Carla Williamson	200
2	S6	Dante Morse	198
3	S7	Kaiser William	219
4	S8	Madeeha Preston	201

7. Write a Pandas program to join the two dataframes with matching records from both sides where available.

Test Data:

student_data1:

	student_id	name	marks
0	S1	Danniella Fenton	200
1	S2	Ryder Storey	210
2	S3	Bryce Jensen	190
3	S4	Ed Bernal	222

4 S5 Kwame Morin 199

student_data2:

	student_id	name	marks
0	S4	Scarlette Fisher	201
1	S5	Carla Williamson	200
2	S6	Dante Morse	198
3	S7	Kaiser William	219
4	S8	Madeeha Preston	201

8. Write a Pandas program to join (left join) the two dataframes using keys from left dataframe only.

Test Data:

data1:

	key1	key2	P	Q
0	K0	K0	P0	Q0
1	K0	K1	P1	Q1
2	K1	K0	P2	Q2
3	K2	K1	P3	Q3

data2:

	key1	key2	R	S
0	K0	K0	R0	S0
1	K1	K0	R1	S1
2	K1	K0	R2	S2
3	K2	K0	R3	S3

9. Write a Pandas program to join two dataframes using keys from right dataframe only.

Test Data:

data1:

	key1	key2	P	Q
0	K0	K0	P0	Q0
1	K0	K1	P1	Q1

```
2 K1 K0 P2 Q2
3 K2 K1 P3 Q3
```

data2:

```
key1 key2 R S
0 K0 K0 R0 S0
1 K1 K0 R1 S1
2 K1 K0 R2 S2
3 K2 K0 R3 S3
```

10. Write a Pandas program to merge two given datasets using multiple join keys.

Test Data:

data1:

```
key1 key2 P Q
0 K0 K0 P0 Q0
1 K0 K1 P1 Q1
2 K1 K0 P2 Q2
3 K2 K1 P3 Q3
```

data2:

```
key1 key2 R S
0 K0 K0 R0 S0
1 K1 K0 R1 S1
2 K1 K0 R2 S2
3 K2 K0 R3 S3
```

11. Write a Pandas program to create a new DataFrame based on existing series, using specified argument and override the existing columns names.

12. Write a Pandas program to create a combination from two dataframes where a column id combination appears more than once in both dataframes.

Test Data:

data1:

	key1	key2	P	Q
0	K0	K0	P0	Q0
1	K0	K1	P1	Q1
2	K1	K0	P2	Q2
3	K2	K1	P3	Q3

data2:

	key1	key2	R	S
0	K0	K0	R0	S0
1	K1	K0	R1	S1
2	K1	K0	R2	S2
3	K2	K0	R3	S3

13. Write a Pandas program to combine the columns of two potentially differently-indexed DataFrames into a single result DataFrame.

Test Data:

data1:

	A	B
K0	A0	B0
K1	A1	B1
K2	A2	B2

data2:

	C	D
K0	C0	D0
K2	C2	D2
K3	C3	D3

14. Write a Pandas program to merge two given dataframes with different columns.

Test Data:

data1:

	key1	key2	P	Q
--	------	------	---	---

```
0 K0 K0 P0 Q0
1 K0 K1 P1 Q1
2 K1 K0 P2 Q2
3 K2 K1 P3 Q3
```

data2:

```
   key1 key2  R  S
0  K0   K0 R0 S0
1  K1   K0 R1 S1
2  K1   K0 R2 S2
3  K2   K0 R3 S3
```

15. Write a Pandas program to Combine two DataFrame objects by filling null values in one DataFrame with non-null values from other DataFrame.

Test Data:

Original DataFrames:

```
   A  B
0 NaN 3
1 0.0 4
2 NaN 5
```

```
   A  B
0 1 3.0
1 1 NaN
2 3 3.0
```